

Unit 4

The matrix inverse.

Unit Goals

- Be able to explain what is meant by the inverse of a square matrix.
- Be able to explain the relationship between the singularity of a square matrix and the existence of a unique solution to a linear system.
- Be able to explain the relationship between elementary row operations and elementary matrices.
- Be able to use Gauss-Jordan elimination to invert a matrix or demonstrate that the inverse does not exist.
- Be able solve real-world problems with linear models using the matrix inverse.

Unit 4.1

What is a matrix inverse?

Identity matrices.

- Recall that we can define a set of square identity matrices, I :

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & & 0 & 0 \\ & \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

- Multiplication of any matrix by appropriately-sized identity matrix gives the same matrix as the result:

$$I_m A = A I_n = A, \forall A \in \mathbb{R}^{m \times n}$$

Multiplicative identities

- When A is square, we can write:

$$IA = AI = A, \forall A \in \mathbb{R}^{n \times n}$$

- Formally: I_n is the multiplicative identity element of the ring of all n -square matrices.
- I is analogous to the number 1 in the ring of integers, the fields of **integer**, **rational** and **real** numbers, etc.

$$1x = x1 = x, \forall x \in \mathbb{Z}, \mathbb{Q}, \mathbb{R}$$

Multiplicative inverses

- All real numbers other than zero have a multiplicative inverse, a number we can multiply by to get the multiplicative identity:

$$\forall x \in \mathbb{Q}, \mathbb{R}, x \neq 0, \exists y \in \mathbb{Q}, \mathbb{R}: xy = yx = 1, y \equiv x^{-1} = 1/x$$

- We say that the real (and the rational) numbers form a field because of this.
- The multiplicative inverse of an n -square matrix would be defined:

$$\text{Given } A \in \mathbb{R}^{n \times n}, \text{ if } \exists B \in \mathbb{R}^{n \times n}: AB = BA = I, \text{ then } B \equiv A^{-1}$$

Multiplicative inverses

Given $A \in \mathbb{R}^{n \times n}$, if $\exists B \in \mathbb{R}^{n \times n}: AB = BA = I$, then $B \equiv A^{-1}$

Questions:

- For a given A , is there a matrix B such that $AB = I$?
- If $AB = I$, must $BA = I$?
- How can we find this matrix?
- Does such a matrix exist for all n -square matrices?

Existence and uniqueness of the inverse.

- **Not all square matrices have an inverse.**
- **If a matrix has an inverse, it is unique.**
- **If a square matrix DOES have an inverse, we say that it is invertible or non-singular.**
- **If a square matrix DOES NOT have an inverse, we say that it is non-invertible or singular.**

Properties of the matrix inverse

- If an n -square matrix, A , is non-singular, then:

$$AA^{-1} = A^{-1}A = I \leftrightarrow (A^{-1})^{-1} = A$$

$$(A^T)^{-1} = (A^{-1})^T$$

- If n -square matrices, A and B , are both non-singular, then AB and BA are also non-singular and:

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (BA)^{-1} = A^{-1}B^{-1}$$

- Remember, most likely $AB \neq BA$.
- If either A or B (or both) are singular, then AB and BA are also singular.

Matrix inverses and linear systems

A linear system of equations can be written as:

$$A\vec{x} = \vec{b}$$

- When the number of equations is the same as the number of variables, A is square, and $\vec{x}$ and $\vec{b}$ are the same length.
- If A has an inverse:

$$A\vec{x} = \vec{b} \rightarrow A^{-1}A\vec{x} = A^{-1}\vec{b} \rightarrow \vec{x} = A^{-1}\vec{b}$$

- A solution is given by $\vec{x} = A^{-1}\vec{b}$!
 - Note: A is the coefficient matrix, NOT the augmented matrix.

Non-singularity and linear systems

- A square coefficient matrix implies that the number of equations (rows) exactly equals the number of variables (columns); such systems *often* have a single unique solution:
 - There will be a unique solution unless Gaussian elimination creates **a zero row in the coefficient matrix**;
 - Whether this leads to many solutions or to none depends on the right-hand side (the last column of the augmented matrix).

Non-singularity and linear systems

- If a system has a unique solution for one right-hand side, it will have a unique solution for every right-hand side.
- If the coefficient matrix is **non-singular**, we can find a solution for every right-hand side ($\vec{x} = A^{-1}\vec{b}$) $\rightarrow$ **the solution must be unique!**
- If the coefficient matrix is **singular**, then the system has **either no solution or many**, depending on the right-hand side.

Utility of the inverse: multiple right-hand sides.

- Many applications involve a need to solve the same linear system with many different right-hand sides.

$$A\vec{x} = \vec{b}_i, i = 1, 2, \dots, N$$

- We could use Gaussian elimination with a combined augmented matrix:

$$\begin{bmatrix} A & \vec{b}_1 & \cdots & \vec{b}_N \end{bmatrix} \rightarrow \begin{bmatrix} U & \vec{b}_1' & \cdots & \vec{b}_N' \end{bmatrix} \rightarrow \begin{bmatrix} R & \vec{x}_1 & \cdots & \vec{x}_N \end{bmatrix}$$

- But we may not know all the right-hand sides at the same time!
 - Example: Industrial production in the face of dynamic consumer demand.
- If A has an inverse, we can use that to solve any right-hand side:

$$\vec{x}_i = A^{-1}\vec{b}_i$$

- This is just matrix-vector multiplication.

Utility of the inverse: How does a change in the RHS change the solution?

- Many applications involve uncertainty in values. Example: industrial production targets are selected by *predicted* demand.

- How does the solution change for a given change in the RHS?

$$A\vec{x} = \vec{b}, \quad A(\vec{x} + \Delta\vec{x}) = \vec{b} + \Delta\vec{b}$$

- If $\Delta\vec{b}$ is fairly small, is $\Delta\vec{x}$ also small, or is it much larger?

$$\vec{x} = A^{-1}\vec{b}, \quad \vec{x} + \Delta\vec{x} = A^{-1}(\vec{b} + \Delta\vec{b}) = A^{-1}\vec{b} + A^{-1}\Delta\vec{b}$$

$$\vec{x} + \Delta\vec{x} = \vec{x} + A^{-1}\Delta\vec{b} \rightarrow \Delta\vec{x} = A^{-1}\Delta\vec{b}$$

- The matrix inverse transforms a difference in the right-hand side into a difference in the solution!

Unit 4.2

Gauss-Jordan Elimination and the matrix inverse

Non-singularity and linear systems

- For a system of n linear equations in n unknowns:

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{n \times n}$$

- If A is non-singular, then the system has a unique solution given by:

$$\vec{x} = A^{-1}\vec{b}$$

- If the coefficient matrix is singular, then the system has either no solution or many, depending on the right-hand side.
- We can use Gaussian elimination to solve for unique solutions, and to determine whether the solution has no solution or many.
- **What is the relationship between Gaussian Elimination and the matrix inverse?**

Gaussian (Gauss-Jordan) Elimination

For a system of m linear equations in n unknowns, we construct an $m \times (n + 1)$ augmented matrix, $M = [A \quad \vec{b}]$:

- If forward elimination gives any rows that have zeros everywhere except the last column of the augmented matrix, then there is **no solution**.
- If forward elimination gives **less than n non-zero rows**, then some variables must be free variables: there are **many (infinite) solutions**; the augmented matrix after back substitution allows us to write a form for these solutions.
- If forward elimination gives **exactly n non-zero rows**, then every variable is a pivot and there is exactly **one (unique) solution**; the last column of the augmented matrix *after back substitution* is the answer.

Singular systems lead to all-zero rows (in coefficient block).

For a square system of n linear equations in n unknowns, working with $M = [A \quad \vec{b}]$:

- **No solution (singular)**: Forward elimination gives at least one row with zeros everywhere in the coefficient block (non-zero only in last column).
- **Many solutions (singular)**: Forward elimination gives at least one row with zeros everywhere in the coefficient block (less than n non-zero rows).
- **One unique solution (non-singular)**: Forward elimination does not give any zero rows in the coefficient block (exactly n non-zero rows).
- The existence of a zero row in the coefficient block of a row echelon form is both necessary and sufficient for classifying the system as singular.

Non-singular systems lead to the identity matrix.

For a **non-singular** square system of n linear equations in n unknowns, working with $M = [A \quad \vec{b}]$:

- There are exactly n non-zero rows in any echelon form.
- Every variable is a pivot.
- There are no free variables.
- **Back substitution leads to a reduced row echelon form with ones on the diagonal and zeros everywhere else, except for the last column.**
- **The coefficient block of the reduced row echelon form of non-singular system is the $n \times n$ identity matrix!**

Elementary matrices.

- Each step we take in Gaussian Elimination can be expressed in terms of only three elementary row operations:
 - **Addition of a scalar multiple:** $R_i + kR_j \rightarrow R_i$
 - **Scalar multiplication:** $kR_i \rightarrow R_i$
 - **Row exchange:** $R_i \leftrightarrow R_j$
- Each of these operations can be achieved through matrix multiplication by an elementary matrix:
 - The elementary matrices are generated by applying the same operation to the identity matrix.
 - Three classes of row operation lead to three classes of elementary matrix.

Interlude: Block matrices.

- Much of the discussion moving forward is simplified by the idea of block matrices: matrices whose elements are other matrices.
- The blocks need to be sized appropriately to make sense. An example:

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, A \in \mathbb{R}^{p \times r}, B \in \mathbb{R}^{p \times s}, C \in \mathbb{R}^{q \times r}, D \in \mathbb{R}^{q \times s}, M \in \mathbb{R}^{(p+q) \times (r+s)}$$

- The augmented matrix is a block matrix, $M = [A \quad \vec{b}]$; the first block is the coefficient matrix and the second is the RHS vector (both have the same number of rows).

Interlude: Block matrices.

- We can perform operations with block matrices much as we can with normal matrices, as long as the blocks are all sized appropriately. An example:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} (AE + BG) & (AF + BH) \\ (CE + DG) & (CF + DH) \end{bmatrix}$$

- The number of **columns** in A and C must match the number of **rows** in E and F .
- The number of **columns** in B and D must match the number of **rows** in G and H .

Elementary matrices.

- Consider the block matrix:

$$M = [A \quad I]$$

- Multiplication by a matrix, E , gives:

$$EM = E[A \quad I] = [EA \quad EI] = [EA \quad E]$$

- Applying some row operation to M must apply the same rule to A and I (each row of M contains the corresponding rows of A and I):

$$R(M) = R([A \quad I]) = [R(A) \quad R(I)]$$

- If EA gives the same result as $R(A)$ then $E = R(I)$.
 - The elementary matrix for any row operation is generated by applying the same operation to the identity matrix.

Example elementary matrices.

- Row exchange:

$$R_i \leftrightarrow R_j$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = A'$$

Example elementary matrices.

- **Addition of a scalar multiple:** $R_i + kR_j \rightarrow R_i$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

Example elementary matrices.

- **Scalar multiplication:**

$$kR_i \rightarrow R_i$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{1/2 R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} \xrightarrow{1/2 R_2 \rightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1/2 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1/2 \\ 0 & -1 & -3 \end{bmatrix} = A'$$

Gaussian Elimination with matrices.

- We start with a sequence of r row operations, forward elimination, that leads to a row echelon (upper triangular) form:

$$M \xrightarrow{R_{f1}} M_{f1} \xrightarrow{R_{f2}} M_{f2} \xrightarrow{R_{f3}} \cdots \xrightarrow{R_{fr-1}} M_{fr-1} \xrightarrow{R_{fr}} M_{fr} = U$$

- Each operation can be achieved by multiplication by an elementary matrix:

$$M \rightarrow E_{f1}M \rightarrow E_{f2}(E_{f1}M) \rightarrow \cdots \rightarrow E_{fr} \left(E_{fr-1} \left(\cdots E_{f2}(E_{f1}M) \right) \right) = U$$

- Forward elimination becomes a product of elementary matrices!

$$E_{fr} \left(E_{fr-1} \left(\cdots E_{f2}(E_{f1}M) \right) \right) = \left(\prod_{i=0}^{r-1} E_{fr-i} \right) M = FM = U, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right)$$

Gaussian Elimination with matrices.

- We continue with a sequence of s row operations, back substitution, that leads to the reduced row echelon form:

$$U \xrightarrow{R_{b1}} U_{b1} \xrightarrow{R_{b2}} U_{b2} \xrightarrow{R_{b3}} \cdots \xrightarrow{R_{bs-1}} U_{bs-1} \xrightarrow{R_{bs}} U_{bs} = R$$

- Again, each operation can be achieved by multiplication by an elementary matrix:

$$U \rightarrow E_{b1}U \rightarrow E_{b2}(E_{b1}U) \rightarrow \cdots \rightarrow E_{bs} \left(E_{bs-1}(\cdots E_{b2}(E_{b1}U)) \right) = R$$

- Back substitution also becomes a product of elementary matrices!

$$E_{bs} \left(E_{bs-1}(\cdots E_{b2}(E_{b1}U)) \right) = \left(\prod_{i=0}^{s-1} E_{bs-i} \right) U = BU = R, \quad B = \left(\prod_{i=0}^{s-1} E_{bs-i} \right)$$

Gaussian elimination is just a matrix product!

$$M \rightarrow \cdots \rightarrow U \rightarrow \cdots \rightarrow R$$

$$U = FM, \quad F = \left(\prod_{i=0}^{r-1} E_{fr-i} \right)$$

$$R = BU, \quad B = \left(\prod_{i=0}^{s-1} E_{bs-i} \right)$$

$$R = B(FM) = (BF)M$$

Gaussian Elimination with matrices.

- If an augmented matrix of a non-singular square system, the reduced row echelon has the identity matrix in the coefficient block and the solution in the last column:

$$M = [A \quad \vec{b}], \quad R = [I \quad \vec{x}]$$

$$R = (BF)M$$

$$[I \quad \vec{x}] = (BF)[A \quad \vec{b}] = [(BF)A \quad (BF)\vec{b}]$$

$$I = (BF)A, \quad \vec{x} = (BF)\vec{b}$$

$$A^{-1}A = I, \quad \vec{x} = A^{-1}\vec{b}$$

- (BF) is the inverse of A !

Matrix inverse by Gauss-Jordan Elimination.

- If we work with a different block matrix, something remarkable happens.

Consider: $M = [A \quad I]$

After full elimination: $R = (BF)M = [(BF)A \quad (BF)I]$

But: $(BF)A = I, (BF)I = BF = A^{-1}$

$$[A \quad I] \xrightarrow{\text{r steps of Forward Elimination}} [U \quad X] \xrightarrow{\text{s steps of back substitution}} [I \quad A^{-1}]$$

Matrix inverse by Gauss-Jordan Elimination.

$$[A \quad I] \xrightarrow{r \text{ steps of Forward Elimination}} [U \quad X] \xrightarrow{s \text{ steps of back substitution}} [I \quad A^{-1}]$$

- Starting with a left block of A and a right block of I , transform through forward elimination and back substitution to obtain a left block of I . The right block now contains A^{-1} .
- If a zero row ever appears in the left block, then A is singular and has no inverse.
- **Note 1:** A is the coefficient matrix, NOT the augmented matrix.
- **Note 2:** A must be square and I must be the same size.

What is a singular matrix?

“Most” square matrices are non-singular; why?

Row picture:

- If two rows are identical or related by a scalar multiple, the matrix must be singular.
- If one row is a linear combination of other rows, the matrix must be singular.
- If any row is zero, the matrix must be singular.

What is a singular matrix?

“Most” square matrices are non-singular; why?

Column picture:

- If two columns are identical or related by a scalar multiple, the matrix must be singular.
- If one column is a linear combination of the others, the matrix must be singular.
- If any column is zero, the matrix must be singular.

Unit 4.2.1

Worked examples computing matrix inverses.

Find the inverse.

- Find the inverse of the following non-singular matrices:

$$(a) \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the inverse.

$$(a) A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 3/2 R_1 \rightarrow R_2} \begin{bmatrix} 2 & 1 & 1 & 0 \\ 0 & 1/2 & -3/2 & 1 \end{bmatrix}$$

$$\xrightarrow{2R_2 \rightarrow R_2} \begin{bmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{bmatrix} \xrightarrow{R_1 - R_2 \rightarrow R_1} \begin{bmatrix} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{bmatrix}$$

$$\xrightarrow{1/2 R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse.

$$(b) A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{aligned} [A \quad I] &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \rightarrow R_2 \\ R_3 - R_1 \rightarrow R_3}} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{bmatrix} \\ &\xrightarrow{R_3 - R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{\substack{R_1 - R_3 \rightarrow R_1 \\ R_2 - R_3 \rightarrow R_2}} \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \\ &\xrightarrow{R_1 - R_2 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} = [I \quad A^{-1}] \end{aligned}$$

Find the inverse.

$$(c) A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 3 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_1 - 4R_4 \rightarrow R_1 \\ R_2 - 3R_4 \rightarrow R_2 \\ R_3 - 2R_4 \rightarrow R_3}} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 & 0 & -4 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{\substack{R_1 - 3R_3 \rightarrow R_1 \\ R_2 - 2R_3 \rightarrow R_2}} \begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse, if possible.

- Find the inverse of the following matrices or explain why the inverse can not exist.

$$(a) \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 0 & -2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & -1 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 2 & 2 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & -2 & 1 & -2 & 1 \\ 1 & 3 & 5 & 7 & 9 \\ 8 & 6 & 4 & 2 & 0 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$$

Find the inverse, if possible.

$$(a) A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 0 & -2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 - 3R_1 \rightarrow R_2 \\ R_3 - 2R_1 \rightarrow R_3}} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -4 & -8 & -3 & 1 & 0 \\ 0 & -4 & -8 & -2 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -4 & -8 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix}$$

- There is no inverse.
- We have encountered a zero row in the coefficient block.

Find the inverse, if possible.

$$(b) A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & -1 & 0 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3+R_1 \rightarrow R_3]{R_2-R_1 \rightarrow R_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$
$$\xrightarrow{R_3-R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -1 & 1 \end{bmatrix}$$

- There is no inverse.
- We have encountered a zero row in the coefficient block.

Find the inverse, if possible.

$$(c) A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} -1 & 1 & -1/2 \\ -1 & 0 & 1 \\ 1 & 0 & -1/2 \end{bmatrix}$$

$$[A \quad I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - 2R_2 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & 0 & 1 \end{bmatrix} \xrightarrow{-1/2 R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix}$$

$$\xrightarrow{\begin{matrix} R_1 - 3R_3 \rightarrow R_1 \\ R_2 - 2R_3 \rightarrow R_2 \end{matrix}} \begin{bmatrix} 1 & 2 & 0 & -3 & 1 & 3/2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix} \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & -1 & 1 & -1/2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1/2 \end{bmatrix} = [I \quad A^{-1}]$$

Find the inverse, if possible.

$$(d) \begin{bmatrix} 1 & -2 & 1 & -2 & 1 \\ 1 & 3 & 5 & 7 & 9 \\ 8 & 6 & 4 & 2 & 0 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$$

- There is no inverse.
- This is a 4×5 matrix; the inverse can only exist for square matrices.

Identify the source of singularity.

- Without using Gauss-Jordan elimination, explain why the following matrices are singular:

$$(a) \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 3 \\ 1 & 0 & 2 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & -2 & 1 & -2 & 1 \\ -2 & 1 & -2 & 1 & -2 \\ 3 & 2 & 1 & 2 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & -4 & 5 & -4 & 3 \end{bmatrix}$$

Identify the source of singularity.

$$(a) \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 3 \\ 1 & 0 & 2 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & -2 & 1 & -2 & 1 \\ -2 & 1 & -2 & 1 & -2 \\ 3 & 2 & 1 & 2 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & -4 & 5 & -4 & 3 \end{bmatrix}$$

(a) The first and third columns are identical, and the third row is simply twice the first.

(b) The second column is all zeros.

(c) The first and fifth column are identical, as are the second and ~~third~~ **fourth** columns.

Unit 4.2.2

Worked examples solving real-world problems with matrix inverses.

Economic equilibrium.

- A coarse 粗糙的 model of a national economy breaks economic activity into three sectors:
 - primary (agriculture and natural resource extraction);
 - secondary (manufacturing); and
 - tertiary (services).
- Consider a “balanced” economy, where to the production of \$1M of output from any sector requires \$0.3M of input from each of the three sectors.

Economic equilibrium.

- If there is an additional (e.g. consumer) **demand for \$100M** from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is **\$150M in demand for primary and secondary sector output**, but **\$200M in demand for the output of the tertiary sector**, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- \$10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

Economic equilibrium.

- **We have multiple different consumer demands to consider, so we should formulate this problem using symbolic constants rather than numbers!**

Economic equilibrium.

- Let x_i and d_i be the total supply and demand, respectively (in millions of dollars), by economic sector i .
- Let b_i be the consumer demand (in millions of dollars) for the product of economic sector i .
- Let the sectors be defined as: agriculture & natural resources $i=1$; manufacturing $i=2$; and services $i=3$.

$$\begin{aligned}d_1 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_1 \\d_2 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_2 \\d_3 &= 0.3x_1 + 0.3x_2 + 0.3x_3 + b_3\end{aligned}$$

- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.

$$d_i = x_i, \forall i$$

$$\vec{x} = A\vec{x} + \vec{b}, A = \begin{bmatrix} 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \\ 0.3 & 0.3 & 0.3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$M = (I - A) = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}$$

$$M\vec{x} = \vec{b}$$

$$\vec{x} = M^{-1}\vec{b}$$

- Compute the inverse.

Economic equilibrium.

$$[M \quad I] = \begin{bmatrix} +0.7 & -0.3 & -0.3 & 1 & 0 & 0 \\ -0.3 & +0.7 & -0.3 & 0 & 1 & 0 \\ -0.3 & -0.3 & +0.7 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{10R_1 \rightarrow R_1 \\ 10R_2 \rightarrow R_2 \\ 10R_3 \rightarrow R_3}} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ -3 & +7 & -3 & 0 & 10 & 0 \\ -3 & -3 & +7 & 0 & 0 & 10 \end{bmatrix}$$

$$\xrightarrow{\substack{R_2 + 3/7 R_1 \rightarrow R_2 \\ R_3 + 3/7 R_1 \rightarrow R_3}} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & \frac{40}{7} & \frac{-30}{7} & \frac{30}{7} & 10 & 0 \\ 0 & \frac{-30}{7} & \frac{40}{7} & \frac{30}{7} & 0 & 10 \end{bmatrix} \xrightarrow{\substack{7/10 R_2 \rightarrow R_2 \\ 7/10 R_3 \rightarrow R_3}} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & -3 & 4 & 3 & 0 & 7 \end{bmatrix}$$

$$\xrightarrow{R_3 + 3/4 R_2 \rightarrow R_3} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & \frac{7}{4} & \frac{21}{4} & \frac{21}{4} & 7 \end{bmatrix} = [U \quad X]$$

Economic equilibrium.

$$[U \quad X] = \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & \frac{7}{4} & \frac{21}{4} & \frac{21}{4} & 7 \end{bmatrix} \xrightarrow{4/7 R_3 \rightarrow R_3} \begin{bmatrix} +7 & -3 & -3 & 10 & 0 & 0 \\ 0 & 4 & -3 & 3 & 7 & 0 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix}$$

$$\xrightarrow{\substack{R_1 + 3R_3 \rightarrow R_1 \\ R_2 + 3R_3 \rightarrow R_2}} \begin{bmatrix} +7 & -3 & 0 & 19 & 9 & 12 \\ 0 & 4 & 0 & 12 & 16 & 12 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} \xrightarrow{1/4 R_2 \rightarrow R_2} \begin{bmatrix} +7 & -3 & 0 & 19 & 9 & 12 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix}$$

$$\xrightarrow{R_1 + 3R_2 \rightarrow R_1} \begin{bmatrix} +7 & 0 & 0 & 28 & 21 & 21 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} \xrightarrow{1/7 R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & 0 & 4 & 3 & 3 \\ 0 & 1 & 0 & 3 & 4 & 3 \\ 0 & 0 & 1 & 3 & 3 & 4 \end{bmatrix} = [I \quad M^{-1}]$$

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for \$100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is \$150M in demand for primary and secondary sector output, but \$200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- \$10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M\vec{x} = \vec{b}, M = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = M^{-1}\vec{b}, M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix} = \begin{bmatrix} 1000 \\ 1000 \\ 1000 \end{bmatrix}$$

- Each sector must produce \$1B of output.

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for \$100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is \$150M in demand for primary and secondary sector output, but \$200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- \$10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M\vec{x} = \vec{b}, M = \begin{bmatrix} +0.7 & -0.3 & -0.3 \\ -0.3 & +0.7 & -0.3 \\ -0.3 & -0.3 & +0.7 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\vec{x} = M^{-1}\vec{b}, M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 150 \\ 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 1650 \\ 1650 \\ 1700 \end{bmatrix}$$

- The primary and secondary sectors must produce \$1.65B of output, while the tertiary must produce \$1.7B.

Economic equilibrium.

- If there is an additional (e.g. consumer) demand for \$100M from each sector, what must the total output from each sector be such the the total supply perfectly matches the total demand in every sector.
- If there is \$150M in demand for primary and secondary sector output, but \$200M in demand for the output of the tertiary sector, what must the total output from each sector be at equilibrium?
- If there the consumer demand values have an uncertainty of +/- \$10M, what is the uncertainty in the total output for each sector needed to achieve equilibrium?

$$M^{-1} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

$$\begin{aligned} M(\vec{x} + \Delta\vec{x}) &= (\vec{b} + \Delta\vec{b}) \\ (\vec{x} + \Delta\vec{x}) &= M^{-1}(\vec{b} + \Delta\vec{b}) \\ \Delta\vec{x} &= M^{-1}\Delta\vec{b} \end{aligned}$$

$$\Delta\vec{x} = \begin{bmatrix} 4 & 3 & 3 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix}$$

- Uncertainties of +/- \$10M in consumer demand lead to uncertainties of +/- \$100M in total equilibrium supply and demand.